SILICONE 3D PRINTERS

By Tasnim Amara

REQUIREMENTS

Requirement	Description
biocompatible	Must print with FDA- and ISO 10993-certified, medically approved silicone
color capability	MUST support colors for skin tone and colored hearing aids.
percision & resolution	Sub-millimeter accuracy (ideally <50 µm) for complex ear canal geometries and thin walls.
scalibility	Capable of batch production and rapid turnaround to meet current and future demand.

1. LYNXTER S300X ($100,000 CAD)

PROS:
- PRINTS WITH CERTIFIED MEDICAL-GRADE SILICONES (RTV2 MEDICAL SILICONES (5,10, 25, 40 SHA), LSR'S)
- DUAL EXTRUSION
- HIGH PRECISION (12.5 UM XY RESOLUTION)
- LARGE BUILD VOLUME FOR BATCH MANUFACTURING

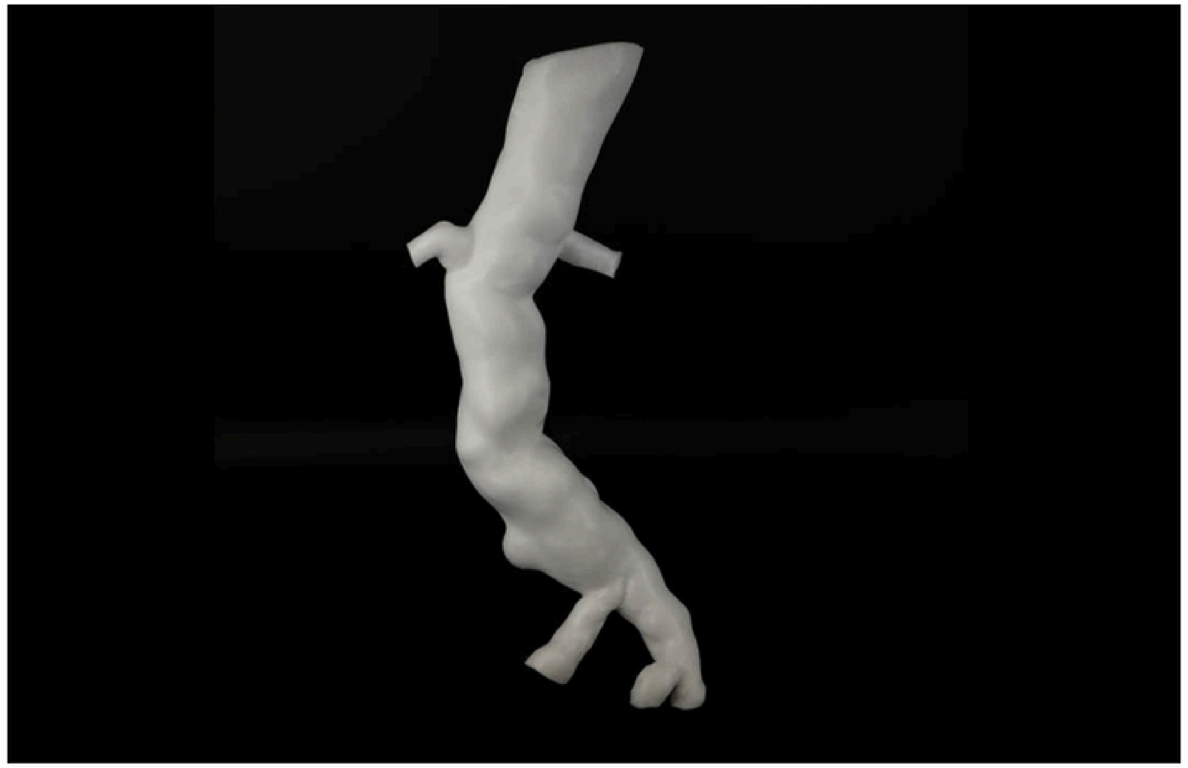
CONS:
- TO ACHEIVE COLOR, WE MUST DYE THE SILICONE USING A HIGH SPEED MIXER

MATERIAL COMPATIBILITY: MEDICAL-GRADE, TWO-PART SILICONES, MOST MANUFACTURERS GOOD (ELKEM, NUSIL, COP CHIME)

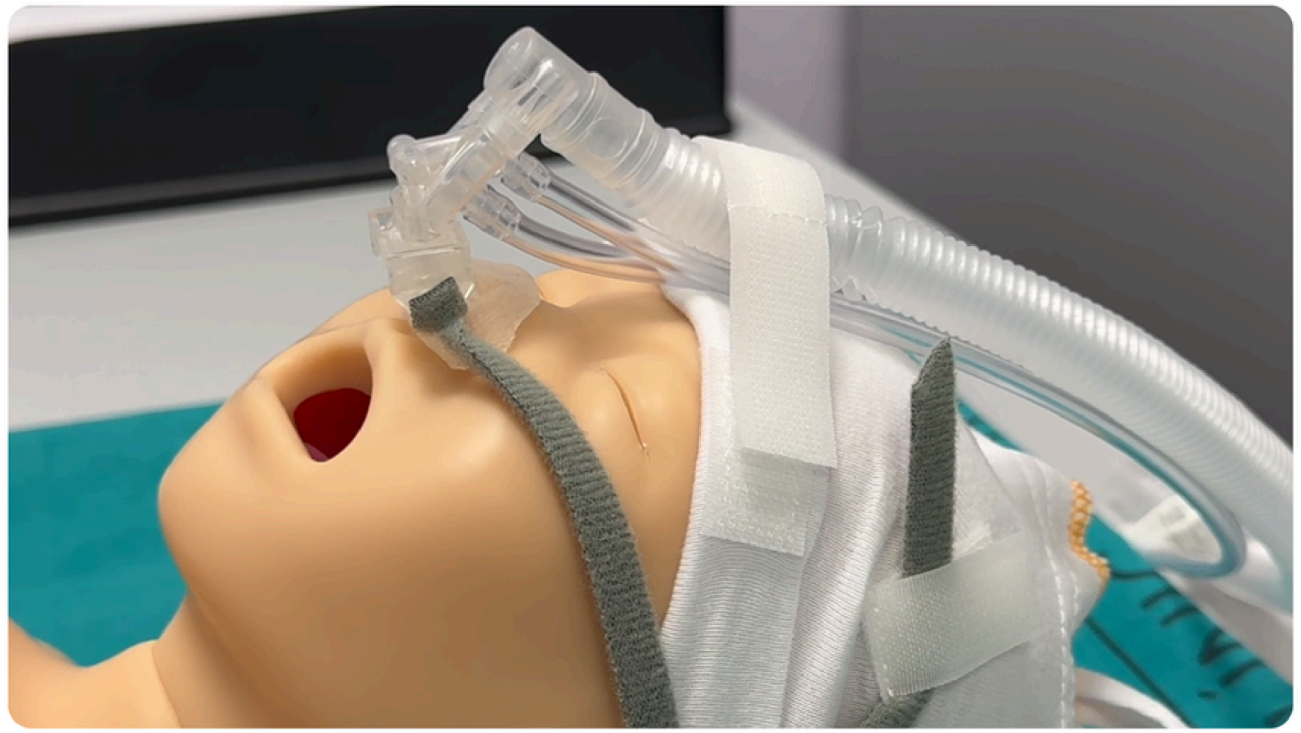
CONSIDER: SILK SKIN?

1. LYNXTER S300X SAMPLES

Ear epithet- Silicone COPSIL 10shA



Medical aorta - Silicone LSR Dynasoft

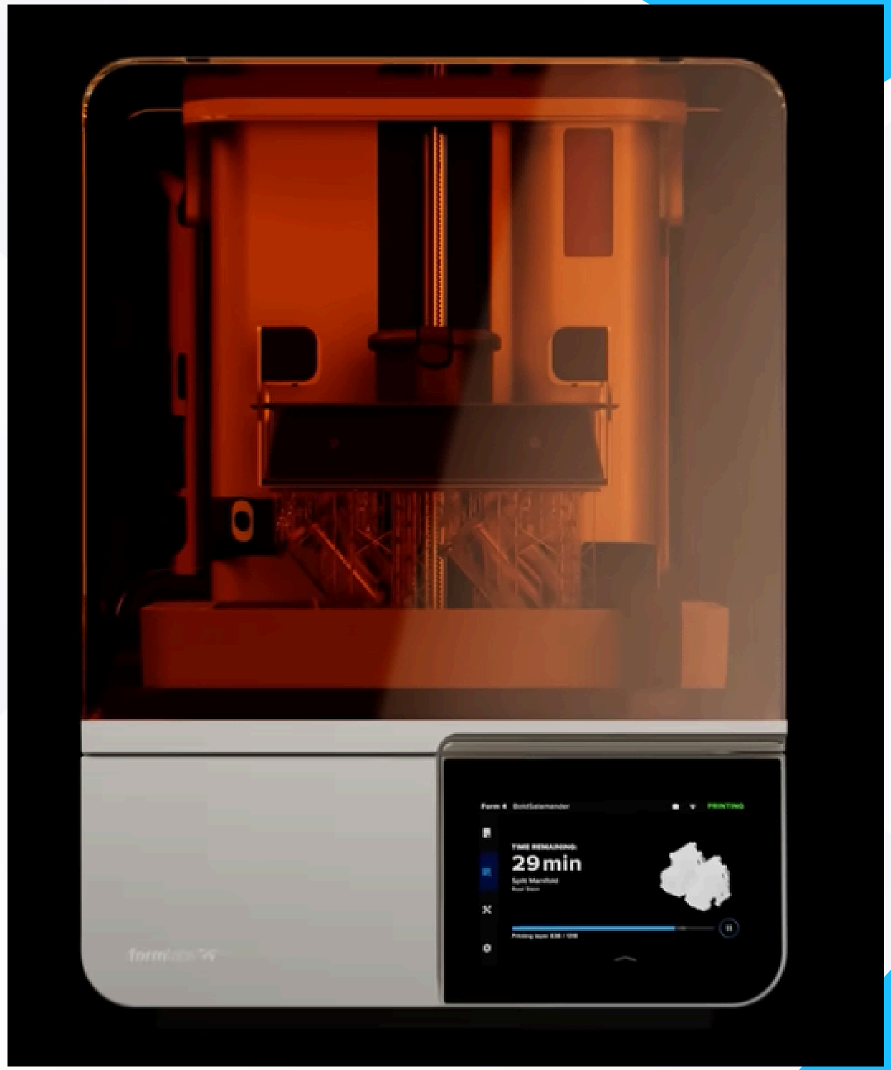


Custom 3D-printed silicone mask

2. FORMLABS FORM 4B WITH SILICONE 40A RESIN ($12,006)

PROS:

- HIGH-PRECISION 50UM XY RESOLUTION FOR SMOOTH, DETAILED SURFACES AND PERFECT ANATOMICAL FIT.
- CERTIFIED BIOCOMPATIBLE MATERIALS WITH SILICONE 40A RESIN FOR SAFE, STERILIZABLE DEVICES.
- FAST LFD PRINT TECHNOLOGY DELIVERING 4X THE THROUGHPUT OF ITS PREDECESSOR.
- LARGE 15 X 16.8 X 19.7 CM BUILD VOLUME FOR HIGH-CAPACITY BATCH PRODUCTION.



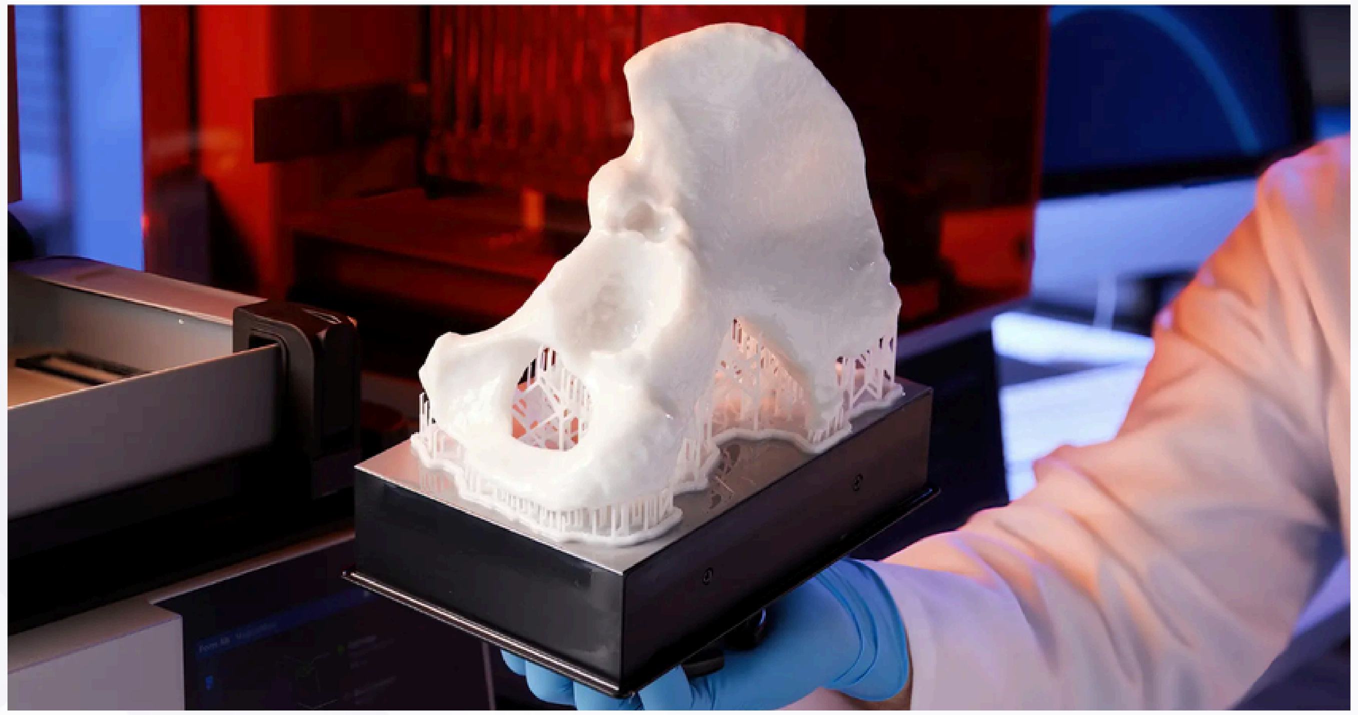
CONS:

- SILICONE 40A RESIN PENDING FULL ISO 10993 CERTIFICATION FOR LONG-TERM SKIN CONTACT
- LIMITED TO FORMLABS BRAND MATERIALS FOR THE MOST PART

MATERIAL COMPATIBILITY: SILICONE 40A RESIN, OPEN MATERIAL LICENSE TO GET 3D PARTY MATERIALS

COLOR SOLUTION: HIGHSPEED MIXING

2. FORMLABS 4B SAMPLES



3. SPECTROPLAST SAM

PROS:

- PRINTS WITH 100% PURE, MEDICAL-GRADE SILICONES (SHORE 20A-60A) FOR BIOCOMPATIBILITY
- 100MM RESOLUTION DELIVERS ISOTROPIC MATERIAL PROPERTIES AND SMOOTH SURFACES
- PROVIDES AN ON-DEMAND PRINTING SERVICE, ELIMINATING CAPITAL EXPENDITURE

CONS:

- NO PRINTER TO PURCHASE; YOU OUTSOURCE ALL PRODUCTION, REDUCING IN-HOUSE CONTROL
- THE WEBSITE DOES NOT MENTION MULTI-COLOR PRINTING

MATERIAL COMPATIBILITY: MEDICAL-GRADE SILICONE

COLOR SOLUTION: NOT MENTIONED, MUST ASK MANUFACTURER ABOUT IT

4. CARBON L1 ($170,900/YEAR FOR 3 YEARS)

PROS:
- MEANT FOR HIGH-THROUGHPUT PRODUCTION
- TRUSTED BY TOP BRANDS (ADIDAS) AND DENTAL LABS
- 1000 CM^2 OF BUILD AREA
- GREAT ACCURACY (+- 70 MICROMETERS)
- HAS BEEN USED SPECIFICALLY FOR EAR MOLD USE

CONS:
- SOFTWARE SUBSCRIPTIONS
- NO COLOR OPTIONS
- NO MATERIAL FLEXIBILITY

MATERIAL COMPATIBILITY: CARBON SIL 30

COLOR SOLUTION: SITE HAS SOME DIFFERENT COLOR SHOWN BUT STATES NO COLOR OPTIONS. ELASTOMERS?

4. CARBON L1 SAMPLES

Dental Thermoforming Models

adidas: 4DFWD 2 midsole

5. CARBON M3 MAX ($116,500/YEAR FOR 3 YEARS)

PROS:

- DOUBLE THE PRINT AREA OF THE M3 (307 X 163 X 305 MM) FOR HIGH-CAPACITY BATCH PRODUCTION.

CONS:

- YEARLY CONTRACT

MATERIAL COMPATIBILITY: CARBON SIL 30

COLOR SOLUTION: SITE HAS SOME DIFFERENT COLOR SHOWN BUT STATES NO COLOR OPTIONS. ELASTOMERS?

RECOMMENDED PRINTER: FORMLABS 4B

- GOOD PRICE
- COMPATIBLE WITH DIFFERENT MATERIALS
- HAS MEDICAL OPTIONS
- TRUSTED BY LOTS OF DENTISTS, THATS A GOOD FACTOR

ALTERNATIVE: IF THE PRICE ISNT THAT MUCH OF A FACTOR, LYNXTER S300X IS A GOOD OPTION. YOU DO NOT HAVE TO PAY TO USE OTHER MATERIALS, WHICH COULD BE GOOD IN THE LONG RUN. (HAVENT DONE THE MATH TO FIGURE THAT OUT)

THANK YOU